Project Report

SIMULATION AND MODELING LAB

Course Code: CSE-413

Project: Weather simulation and statistical analysis

SUBMITTED TO

Afroza Ahmed Smirty

Lecturer of University of Information Technology and Sciences (UITS)

SUBMITTED BY

Md. Mefazur Rahman (0432310005101035)
Nazmul Chowdury (0432310005101044)
Md. Alif Reza Shimanto (0432310005101046)
Md. Shoyaif Rahman (0432310005101050)
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TITLE: Weather Simulation & Statistical Analysis Using Python

ABSTRACT
This project presents a simulation-based weather analysis system developed using Python programming language and statistical modeling techniques. The project simulates daily temperature data for two cities, Dhaka and Cox's Bazar, over a period of 30 days using probability distributions. Various statistical hypothesis tests such as the One-Sample t-Test, Two-Sample t-Test, and Chi-Square Test are performed to analyze the generated data. Additionally, animated graphical visualization is used to demonstrate temperature changes over time. The purpose of this project is to demonstrate how simulation and statistical analysis can be applied in real-world weather forecasting and data interpretation.

Chapter 1: INTRODUCTION

1.1 Background
Weather forecasting and climate analysis are important applications of simulation and statistical modeling. Real-world weather systems involve uncertainty, randomness, and dynamic environmental factors. By using simulation techniques, it becomes possible to generate realistic weather data and analyze patterns without collecting real-time datasets.

Statistical tests help researchers determine whether observed weather patterns are meaningful or occur due to random chance. Python provides powerful libraries such as NumPy, SciPy, and Matplotlib which make simulation and visualization easier.

1.2 Problem Statement
Weather conditions vary significantly between different cities. It is often difficult to manually analyze whether differences in temperature are statistically significant. Therefore, this project aims to simulate weather data and apply statistical analysis methods to compare temperatures between cities.

1.3 Objectives
- To generate simulated weather temperature data using probability distributions.
- To perform statistical hypothesis testing on generated data.
- To compare temperature patterns between Dhaka and Cox's Bazar.
- To visualize weather changes using graphs and animations.
- To understand the practical application of simulation and modeling techniques.

1.4 Scope of the Project
This project focuses only on temperature simulation for two cities over 30 days. The system can be expanded in the future to include rainfall prediction, humidity analysis, wind speed simulation, machine learning forecasting, and real-time weather API integration.

Chapter 2: METHODOLOGY

2.1 Tools and Technologies Used

Tool/Technology	Purpose
Python	Main programming language
NumPy	Random data generation and array operations
SciPy	Statistical testing
Matplotlib	Data visualization and animation
Colab Notebook	Development environment

2.2 Simulation Process

The project simulates weather temperature data using the Normal Distribution.

Dhaka Temperature Simulation
- Mean Temperature = 32°C
- Standard Deviation = 3

Cox's Bazar Temperature Simulation
- Mean Temperature = 27°C
- Standard Deviation = 4

2.3 Data Visualization

Line charts are used to represent daily temperature changes. Animation techniques are also applied to dynamically display how temperature changes over time. Visualization helps users understand trends and compare temperature fluctuations between cities.

Chapter 3: STATISTICAL ANALYSIS

3.1 One-Sample t-Test

Purpose

The One-Sample t-Test checks whether the average temperature of Dhaka significantly differs from the national average temperature of 30°C.

Hypothesis

- Null Hypothesis (H0): Mean temperature of Dhaka = 30°C
- Alternative Hypothesis (H1): Mean temperature of Dhaka ≠ 30°C

Interpretation

If the p-value is less than 0.05, the null hypothesis is rejected, meaning Dhaka's average temperature is significantly different from 30°C.

3.2 Two-Sample t-Test

Purpose

The Two-Sample t-Test compares the average temperatures between Dhaka and Cox's Bazar.

Hypothesis

- Null Hypothesis (H0): Both cities have equal average temperatures.
- Alternative Hypothesis (H1): The average temperatures are different.

Interpretation

A small p-value indicates that the temperature difference between the two cities is statistically significant.

3.3 Chi-Square Test

Purpose

The Chi-Square Test checks whether temperature categories are evenly distributed.

Categories Used

- Cool
- Warm
- Hot

Interpretation

If the calculated Chi-Square statistic is large and p-value is small, the distribution is not uniform.

Chapter 4: SYSTEM DESIGN

4.1 Workflow Diagram

- Generate random temperature data
- Store generated values
- Perform statistical tests
- Visualize results using graphs
- Animate temperature changes
- Interpret statistical outputs

4.2 System Architecture

Input Data → Simulation Engine → Statistical Analysis → Visualization → Output Report

Chapter 5: RESULTS AND DISCUSSION

5.1 Generated Data Analysis

The generated temperature data shows realistic variations for both cities. Dhaka generally maintains a higher average temperature than Cox's Bazar. The simulation successfully demonstrates how probability distributions can mimic real-world weather behavior.

5.2 Statistical Findings

Test	Objective	Expected Outcome
One-Sample t-Test	Compare Dhaka mean with 30°C	Significant difference
Two-Sample t-Test	Compare two cities	Dhaka hotter than Cox's Bazar
Chi-Square Test	Analyze category distribution	Uneven category distribution

5.3 Visualization Results

The animated line graph clearly displays daily temperature fluctuations over the 30-day period. Visualization improves understanding of data trends and supports statistical conclusions.

Chapter 6: ADVANTAGES AND LIMITATIONS

6.1 Advantages

- Easy to simulate realistic temperature data
- Demonstrates practical use of statistics
- Interactive visualization improves understanding
- Simple and efficient Python implementation
- Useful for academic learning purposes

6.2 Limitations

- Uses only simulated data
- Limited to temperature analysis
- Small dataset size (30 days)
- Does not include real-time forecasting

Chapter 7: FUTURE IMPROVEMENTS

1. Real-time weather API integration
2. Machine learning prediction models
3. Rainfall and humidity simulation
4. Dashboard-based visualization
5. Database integration
6. Multi-city comparative analysis

Chapter 8: CONCLUSION

This project successfully demonstrates the use of simulation and statistical analysis techniques in weather modeling. By generating synthetic temperature data and applying statistical tests, meaningful insights were obtained regarding weather differences between cities.

The project also highlights the effectiveness of Python libraries in scientific computing, visualization, and hypothesis testing. Overall, the system serves as an excellent academic example of applying simulation concepts to real-world problems.